

The Definition of the Derivative



Using the limit definition

- 1 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 3x^2 - x$.
 - 2 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+2}$.
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The derivative as a limit at a point

- 3 Use $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(3)$ for $f(x) = x^2 - 2x$.
 - 4 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.
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Differentiability and graphical interpretation

- 5 Explain why $f(x) = |x - 2|$ is not differentiable at $x = 2$, computing the one-sided derivatives.
 - 6 State three conditions under which a function fails to be differentiable at a point, with a brief description of each (e.g. corner, cusp, vertical tangent, discontinuity).
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Power, sum, and constant rules

- 7 Differentiate each function using the power rule:
a $f(x) = 7x^5$ **b** $f(x) = \frac{4}{x^2}$ **c** $f(x) = 6\sqrt[3]{x}$ **d** $f(x) = 9$
 - 8 Differentiate $f(x) = 5x^4 - 3x^3 + 2x - 7$.
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Tangent lines from the derivative

- 9 Using the limit definition result $f'(x) = 6x - 1$ found earlier for $f(x) = 3x^2 - x$, find the equation of the tangent line to f at $x = 1$.
 - 10 Find the equation of the tangent line to $f(x) = x^3$ at $x = 2$, using $f'(2) = 12$ (found earlier).
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Estimating derivatives numerically

11 Given $f(1.9) = 3.42$ and $f(2.1) = 3.98$, use a symmetric difference quotient to estimate $f'(2)$.